Volatility-Managed Portfolios:

Replication and Post-Publication Evidence

A replication and extension of Moreira and Muir (2017)

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Abstract

We replicate Moreira and Muir (2017) using publicly available data from the Kenneth French Data Library, extending the original sample through 2025 to test post-publication robustness. We construct volatility-managed versions of the market, value, momentum, profitability, and investment factors by scaling monthly factor returns by the inverse of the previous month's realized variance, then evaluate performance using alpha regressions with Newey-West standard errors. Our results confirm the central mechanism of the original paper: volatility is highly time-varying, while expected returns do not rise proportionally in high-volatility periods, causing conditional Sharpe ratios to deteriorate. The strongest evidence appears for momentum, which earns an annualized alpha of 11.85 percent, and for the market and profitability factors, while value and investment show weaker performance in the extended sample. These findings support volatility management as a useful timing strategy, but also suggest that its benefits are factor-specific and partly affected by post-publication decay.

1. Introduction

1.1 The puzzle

Standard finance theory predicts a positive trade-off between risk and return: periods of elevated volatility should be compensated by elevated expected returns, leading rational investors to take on more risk during turbulent regimes. Yet a long empirical literature, dating back at least to French, Schwert, and Stambaugh (1987), has struggled to identify a robust positive relationship between conditional volatility and conditional expected returns. In some sub-periods the empirical relationship is statistically indistinguishable from zero; in others, as documented by Whitelaw (1994), it is even negative.

This stylized fact creates a puzzle for asset pricing theory. Habit-formation models (Campbell and Cochrane 1999), long-run risk models (Bansal and Yaron 2004), rare-disaster models (Wachter 2013), and intermediary asset pricing models (He and Krishnamurthy 2013) all share a common prediction: risk premia rise in volatile periods, implying that a rational investor should lean into risk when realized volatility is elevated. The empirical evidence does not clearly support this implication. If the conditional Sharpe ratio is not constant — and a substantial body of evidence suggests it falls in high-volatility states — then the optimal mean-variance response is to scale exposure down, not up, when volatility rises.

1.2 Moreira and Muir's findings

Moreira and Muir (2017) formalize and exploit this asymmetry. They construct volatility-managed portfolios that scale exposure to each of eight well-known factor strategies, the market, value, momentum, profitability, investment, return on equity, betting-against-beta, and currency carry, by the inverse of the previous month's realized variance. The scaling constant is calibrated so that the unconditional variance of the managed factor matches that of the original, ensuring that Sharpe-ratio comparisons reflect timing rather than scale. Their results are striking. Across all eight factors, the volatility-managed counterpart produces a large and statistically significant alpha relative to a buy-and-hold position in the original factor. The market factor alone delivers an annualized alpha of approximately 4.86 percent with a Newey-West HAC t-statistic above 3, while momentum delivers an alpha exceeding 12 percent with a t-statistic above 4. Sharpe ratios increase by 50 to 80 percent on average across the eight factors, and a mean-variance investor combining all eight managed factors realizes a certainty-equivalent gain exceeding 125 percent relative to the unmanaged benchmark. The finding directly challenges every leading risk-based asset pricing model and provides academic legitimacy to volatility-targeting strategies already in widespread industry use, including target-volatility exchange-traded funds, risk-parity overlays, and managed-futures strategies.

1.3 Our contribution

In this paper we replicate the central empirical results of Moreira and Muir (2017) using publicly available data from the Kenneth French Data Library, and we extend the original sample by approximately ten years to test post-publication robustness. We focus on the five Fama-French and momentum factors for which daily and monthly returns are directly available from a single source, and we reference the original authors' results for factors that require supplementary datasets.

Our replication yields three principal findings. First, for the market, momentum, and profitability factors, our point estimates lie within roughly half a percentage point of the alphas reported by Moreira and Muir, with momentum delivering a t-statistic above 6 in our full sample. Second, the value and investment factors generate substantially weaker alphas in our extended sample, with the cross-sectional and average investment (CMA) alpha collapsing toward zero, a pattern consistent with the broader literature on post-publication factor decay documented by McLean and Pontiff (2016). Third, a subsample analysis confirms that alpha magnitudes are largest during the most volatile decades of the twentieth and early twenty-first centuries, supporting the conditional Sharpe-ratio mechanism proposed by the original authors. The Python replication code and source data are provided in the supplementary materials and are reproducible from public sources.

1.4 Roadmap

The remainder of the paper is organized as follows. Section 2 reviews the relevant literature on time-varying expected returns and volatility timing. Section 3 develops the theoretical framework underlying the volatility-managed strategy. Section 4 describes our data and methodology. Section 5 presents the main empirical results, including the direct predictability test and the alpha regressions across factors. Section 6 discusses the post-publication evidence, robustness checks, and theoretical implications. Section 7 concludes.

2. Literature Review

We organize the related literature along three dimensions. First, we review the evidence on time-variation in expected stock returns. Second, we summarize the work on volatility forecasting and the conditional mean-volatility relation. Third, we discuss the factor-investing literature and the most direct precursors to the volatility-managed approach. Moreira and Muir's (2017) contribution lies at the intersection of these strands.

2.1 Time-varying expected returns

The empirical literature documenting time-variation in expected stock returns is among the most influential in modern finance. Campbell and Shiller (1988) provide the seminal evidence: the dividend-price ratio, derived from a log-linearization of the present-value identity, forecasts long-horizon real returns on the aggregate stock market. Their key insight is that variation in price-dividend ratios must reflect either variation in expected dividends or variation in expected returns, and the data strongly favor the latter interpretation. Fama and French (1989) extend this evidence to corporate bond returns and document that dividend yields and term spreads contain economically meaningful predictive content at horizons of several years.

Subsequent work has refined and qualified these findings. Lettau and Ludvigson (2001) propose a consumption-wealth ratio (denoted cay) that improves upon traditional valuation ratios for predicting excess equity returns at quarterly horizons. Cochrane (2008, 2011) provides the most systematic case for return predictability, arguing that essentially all variation in aggregate valuation ratios reflects discount-rate news rather than cash-flow news. Yet the magnitude of the predictability remains modest at monthly frequencies: out-of-sample R-squared for return forecasts rarely exceeds three percent at the one-month horizon, and in-sample evidence has been criticized on grounds of selection bias and small-sample inference (Stambaugh 1999; Welch and Goyal 2008). The picture that emerges is one in which expected returns vary over time in economically interesting ways, but the variation is slow-moving and only weakly forecastable from one month to the next.

2.2 Volatility forecasting and factor strategies

In sharp contrast to expected returns, conditional return volatility is highly persistent and easily forecasted. French, Schwert, and Stambaugh (1987) document strong positive autocorrelation in monthly realized volatility and find evidence that conditional volatility is positively related to expected returns within month, although the in-sample relationship proves sensitive to specification. Engle's (1982) Nobel-winning ARCH framework provides a parametric model of conditional variance dynamics, subsequently extended to GARCH (Bollerslev 1986) and EGARCH (Nelson 1991) specifications. Andersen and Bollerslev (1998) show that the simple sum-of-squared-intra-period-returns realized variance estimator delivers nearly equivalent volatility predictability to these parametric filters, without requiring any model assumptions. This is the estimator we adopt in our own replication.

A complementary literature challenges the assumption that the conditional mean-volatility relation is uniformly positive. Glosten, Jagannathan, and Runkle (1993) demonstrate that the response of conditional variance to news is asymmetric: negative shocks induce larger volatility responses than positive shocks of equivalent magnitude. Whitelaw (1994) finds that, in some sub-samples, the conditional mean-volatility relation is actually negative. Brandt and Kang (2004) corroborate this result in a latent-variable VAR framework and document that the unconditional and conditional risk-return relations can have opposite signs. These findings imply that the conditional Sharpe ratio is not constant — an empirical regularity central to the volatility-managed strategy.

Fleming, Kirby, and Ostdiek (2001, 2003) provide the first formal argument that an investor can capture economic value from volatility timing even in the absence of return predictability, by exploiting time-variation in the conditional optimal portfolio weight. Their analysis sets the stage for the synthesis later achieved by Moreira and Muir.

2.3 The synthesis: Moreira and Muir (2017)

The volatility-managed strategy is applied to long-short factor portfolios rather than to the aggregate market alone. The factor literature begins with Fama and French (1992, 1993), who establish that book-to-market and size sort securities into portfolios with statistically distinct mean returns. Carhart (1997) augments this with a momentum factor. Fama and French (2015) propose a five-factor model that adds profitability (RMW) and investment (CMA) factors. Hou, Xue, and Zhang (2015) develop a parallel q-factor model centered on a return-on-equity factor. Frazzini and Pedersen (2014) document the betting-against-beta anomaly, in which low-beta portfolios outperform high-beta portfolios on a risk-adjusted basis; they attribute the result to leverage-constrained investors bidding up high-beta securities. Lustig, Roussanov, and Verdelhan (2011) construct currency carry portfolios that exploit cross-sectional variation in forward discounts.

The direct precursor to the volatility-managed approach is Barroso and Santa-Clara (2015), who demonstrate that scaling momentum exposure by the inverse of recent realized volatility approximately doubles the strategy's Sharpe ratio. Their analysis is restricted to a single factor, but it establishes the empirical proof of concept that subsequent authors generalize. McLean and Pontiff (2016) provide an important caveat that informs our own analysis: many published anomalies experience meaningful post-publication decay as professional investors exploit them. We return to this point in Section 6.

3. Theoretical Framework

3.1 The volatility-managed strategy

Following Moreira and Muir (2017), we construct the volatility-managed counterpart of a return series as a time-varying scaling of the underlying factor return. Let $\mathbf{f_{t+1}}$ denote the excess return of a factor portfolio in month **t+1**, and let $\boldsymbol{\sigma^2_t}$ denote a conditional variance estimate constructed using information available at the end of month **t**. We define the volatility-managed return as:

$$\mathbf{f^\wedge\sigma_{t+1} = (c / \sigma^2_t) \cdot f_{t+1}} \qquad \mathbf{(1)}$$

where **c** is a normalizing constant chosen to ensure that the unconditional variance of the managed return equals that of the original:

$$\mathbf{c^2 = Var(f) / Var(f / \sigma^2)} \qquad \mathbf{(2)}$$

This calibration serves two purposes. First, it ensures that comparisons of Sharpe ratios and other risk-adjusted performance measures between the original and managed series reflect differences in the timing of risk-taking rather than differences in scale. A naïve specification without the constant **c** would either over- or under-lever the strategy depending on the units of $\boldsymbol{\sigma^2}$, rendering economic comparisons across factors meaningless. Second, the calibrated strategy exhibits an average leverage of approximately unity by construction. Although the conditional weight $\mathbf{(c / \sigma^2_t)}$ can substantially exceed one in calm periods and fall well below one in turbulent periods, its long-run average is approximately one. The volatility-managed strategy is therefore not a disguised leverage trade; any improvement in risk-adjusted performance must arise from the timing of risk-taking rather than from a sustained increase in average exposure.

A critical feature of equation **(1)** is the timing convention: $\boldsymbol{\sigma^2_t}$ is constructed from data available through the end of month **t**, while $\mathbf{f_{t+1}}$ is the realized return in month **t+1**. The strategy is therefore implementable in real time without look-ahead bias. Following Andersen and Bollerslev (1998), we define the realized variance estimator as the sum of squared daily returns within month **t**:

$$\mathbf{RV_t = \Sigma_j \in_t r^2_j} \qquad \mathbf{(3)}$$

where $\mathbf{r_j}$ denotes the factor return on day **j**. We require at least 15 daily observations per month for an **RV** estimate to enter the analysis.

3.2 Mean-variance optimality

The volatility-managed strategy admits a natural interpretation in the classical mean-variance framework developed by Markowitz (1952). Consider a single-period investor at date $\mathbf{t}$ who allocates wealth between a risky factor portfolio with excess return $\mathbf{r_{t+1}}$ and a risk-free asset. The investor's preferences over end-of-period wealth are summarised by a mean-variance utility function with risk-aversion parameter $\gamma > 0$,

$$\mathbf{U_t(w_t) = E_t[w_t \cdot r_{t+1}] - (\gamma/2) \cdot Var_t(w_t \cdot r_{t+1})} \qquad \mathbf{(4)}$$

where $\mathbf{w_t}$ denotes the conditional weight placed on the risky asset, $\boldsymbol{\mu_t} = \mathbf{E_t[r_{t+1}]}$ is the conditional expected excess return, and $\boldsymbol{\sigma^2_t} = \mathbf{Var_t(r_{t+1})}$ is the conditional variance.

Expanding the variance term gives:

$$\mathbf{U_t(w_t) = w_t \cdot \boldsymbol{\mu_t} - (\gamma/2) \cdot w^2_t \cdot \boldsymbol{\sigma^2_t}}$$

Differentiating with respect to $\mathbf{w_t}$ yields the first-order condition:

$$\boldsymbol{\mu_t - \gamma \cdot w_t \cdot \sigma^2_t = 0}$$

Solving for $\mathbf{w_t}$ produces the well-known formula for the optimal mean-variance portfolio weight:

$$\mathbf{w^*_t = (1/\gamma) \cdot \boldsymbol{\mu_t} / \boldsymbol{\sigma^2_t}} \qquad \mathbf{(5)}$$

Several features of this optimal weight deserve emphasis. First, the dependence on conditional variance $\boldsymbol{\sigma^2_t}$ in the denominator, rather than on conditional standard deviation $\boldsymbol{\sigma_t}$, is structural rather than incidental. It follows directly from the quadratic form of the risk-penalty term in the mean-variance objective, $\mathbf{(\gamma/2) \cdot w^2 \cdot \boldsymbol{\sigma^2}}$. In any quadratic-utility framework — and in any local Taylor approximation to expected utility for continuously differentiable preferences over wealth the second-moment penalty enters linearly in variance, not in standard deviation. Practitioners are sometimes tempted to interpret optimal weights as proportional to the inverse of conditional standard deviation; this intuition is mathematically incorrect and, when implemented, leads to under-leverage in calm regimes and over-leverage in turbulent regimes relative to the true mean-variance optimum.

Second, the optimal weight $\mathbf{w^*_t}$ is linear in the conditional expected excess return $\boldsymbol{\mu_t}$. This linearity is what permits the substantial economic value of volatility timing in the absence of return predictability. Even if $\boldsymbol{\mu_t}$ is treated as a constant — i.e., even if returns are conditionally unpredictable from the investor's information set — the optimal weight remains non-constant because $\boldsymbol{\sigma^2_t}$ varies through time. Volatility timing is therefore a strategy of exploiting variance dynamics rather than return dynamics, and its economic value does not require the investor to forecast expected returns.

Third, the optimal weight has a transparent decomposition in terms of the conditional Sharpe ratio. Rearranging equation **(5)** using $\mathbf{SR_t} = \boldsymbol{\mu_t} / \boldsymbol{\sigma_t}$,

$$\mathbf{w^*_t = (SR_t / \gamma) \cdot (1 / \sigma_t)}$$

When the conditional Sharpe ratio falls — as it does, empirically, in high-volatility regimes — the optimal weight declines on both margins: directly through $\mathbf{SR_t}$ in the numerator, and through $\boldsymbol{\sigma_t}$ in the denominator. The mean-variance investor responds to volatility shocks by reducing exposure for two distinct reasons.

Comparing equation **(5)** to the volatility-managed construction in equation **(1)**, the connection becomes transparent. Equation **(1)** implements:

$$\mathbf{f^{\sigma}_{t+1} = (c / \sigma^2_t) \cdot f_{t+1}}$$

which has the same functional form as equation **(5)** up to a multiplicative constant. The volatility-managed strategy is therefore a feasible implementation of the optimal mean-variance weight under the working assumption that the conditional expected return $\boldsymbol{\mu_t}$ is approximately constant through time. The calibration constant **c** absorbs both the **(1/γ)** coefficient and any sample-specific scaling required to match unconditional variances; the inverse-variance scaling $\mathbf{1/\sigma^2_t}$ implements the optimal timing rule.

Whether the working assumption — that $\boldsymbol{\mu_t}$ is approximately constant relative to $\boldsymbol{\sigma^2_t}$ — is empirically reasonable is the central econometric question of the paper. If $\boldsymbol{\mu_t}$ and $\boldsymbol{\sigma^2_t}$ move proportionally through time, the ratio $\boldsymbol{\mu_t / \sigma^2_t}$ is constant, the optimal weight is constant, and volatility management adds nothing. If, on the other hand, the dynamics of $\boldsymbol{\mu_t}$ are slow-moving relative to those of $\boldsymbol{\sigma^2_t}$ — as a substantial empirical literature suggests, with traditional return predictors such as the dividend-price ratio capturing only one to three percent of monthly variation in returns while volatility filters routinely capture 30 to 50 percent — then the inverse-variance scaling closely tracks the time-varying mean-variance optimum, and the resulting strategy adds substantial economic value. We assess this empirical question directly in Section 5.

A final note on utility specification is warranted. The mean-variance objective in equation **(4)** is exact under quadratic utility or under joint normality of returns, neither of which is empirically realistic for monthly factor returns. For more general utility specifications, equation **(5)** provides a second-order Taylor approximation to the optimal weight, with higher-order moments such as skewness, kurtosis, and conditional jumps contributing additional corrections. Moreira and Muir (2017) verify in their robustness analysis that their qualitative conclusions are unchanged when higher-order moments are explicitly incorporated. We do not pursue these extensions in our replication, but we note that the economic value of the strategy is robust to plausible departures from strict quadratic-utility assumptions.

3.3 The conditional Sharpe ratio mechanism

The economic case for volatility management rests on a single empirical claim: the conditional Sharpe ratio of factor returns is not constant in time but falls in high-volatility regimes. If a doubling of conditional volatility were accompanied by a proportional doubling of conditional expected return, the conditional Sharpe ratio, defined as the ratio of conditional mean to conditional standard deviation, would be invariant to volatility regimes. In that world, the optimal mean-variance weight in equation **(2)** would also be constant in time, and the volatility-managed strategy of equation **(1)** would deliver no improvement over a buy-and-hold position in the original factor.

In the data, however, the conditional volatility process and the conditional mean process exhibit sharply different degrees of predictability. Conditional volatility is highly persistent: the first-order autocorrelation of monthly realised variance is approximately **0.7** in our sample, and parametric volatility filters such as GARCH **(Bollerslev 1986)** and EGARCH **(Nelson 1991)** deliver similar persistence. Out-of-sample R^2 for one-month-ahead volatility forecasts typically lies in the range of **30 to 50 percent**. Conditional expected returns, by contrast, are forecasted with substantially less precision. The most-studied monthly return predictors, the dividend-price ratio **(Campbell and Shiller 1988)**, the consumption-wealth ratio **(Lettau and Ludvigson 2001)**, and the term spread **(Fama and French 1989)**, yield in-sample R^2 in the range of **one to three percent** at monthly horizons, and out-of-sample performance is often weaker still **(Welch and Goyal 2008)**.

This asymmetry between volatility forecastability and return forecastability implies that, when conditional volatility rises, the conditional expected return rises by substantially less than proportionally. Consequently, the conditional Sharpe ratio falls in high-volatility regimes, and the optimal mean-variance weight declines. Scaling down exposure during high-volatility periods is therefore the rational mean-variance response, exactly the opposite of the prediction made by most structural risk-based asset pricing models, which imply counter-cyclical risk premia and hence counter-cyclical rising optimal exposure. The volatility-managed strategy in equation **(1)** operationalises this insight in a transparent, parameter-free manner. The empirical magnitudes underlying this asymmetry are documented in Section **5**.

3.4 The alpha regression

The economic value of the volatility-managed strategy can be tested directly by asking whether its returns are spanned by a constant exposure to the original factor. If the time-variation in conditional Sharpe ratios documented in Section **3.3** is real, the managed factor should earn a return premium that no static portfolio of the original factor can replicate. Following Moreira and Muir **(2017)**, we formalise this test as a simple time-series regression of the managed factor on its unmanaged counterpart:

$$f^{\sigma}_{t+1} = \alpha + \beta \cdot f_{t+1} + \varepsilon_{t+1} \qquad (3)$$

The intercept α captures the average return of the managed factor that is orthogonal to its loading on the original. A positive and statistically significant α implies that the managed factor cannot be replicated by any static combination of the original factor and the risk-free asset; equivalently, that the mean-variance frontier strictly expands when the managed factor is added to the investment opportunity set.

The slope coefficient **β** captures the average exposure of the managed factor to the original. Under our calibration of **c** in equation **(2)**, **β** is typically below unity and lies in the range of **0.5 to 0.8** across factors. The residual $\boldsymbol{\varepsilon_{t+1}}$ captures the idiosyncratic return of the volatility-managed strategy, the component of **f^σ** that is orthogonal to the original factor and that is the source of any alpha gain.

Throughout the empirical analysis we report Newey and West **(1987)** heteroskedasticity- and autocorrelation-consistent standard errors with a six-month lag truncation, the standard correction for monthly time-series regressions in which residuals may exhibit modest serial correlation. We additionally report the appraisal ratio of the managed factor, defined as the ratio of **α** to the standard deviation of the residual term, annualised by multiplying by the square root of twelve.

The appraisal ratio equals the Sharpe ratio of the long-short portfolio that combines a long position in the managed factor with a short **β**-weighted position in the original. It therefore provides a direct economic measure of the value added by volatility timing relative to a passive position in the original factor. Section **5** reports estimates of **α**, **β**, and the appraisal ratio for each factor in our replication sample.

4. Data and Methodology

4.1 Data sources

We replicate Moreira and Muir (2017) using publicly available factor return data. Our main sample consists of daily and monthly returns for the Fama-French factors and the momentum factor from the Kenneth French Data Library. Specifically, we use the market excess return, SMB, HML, RMW, CMA, and MOM factors, which allow us to construct both monthly factor returns and monthly realized variance measures from daily data. The use of daily and monthly factor returns follows the data structure in Moreira and Muir (2017), who construct volatility-managed portfolios by scaling monthly returns using the previous month's realized variance. Our sample period is determined by the availability of daily factor data: the market, SMB, HML, and MOM factors begin in 1926, while RMW and CMA begin in 1963.

Table 1. Data sources and sample periods.

Factor	*Description*	*Source*	*Sample start*	*Frequency*
MKT	Market excess return	Kenneth French Data Library	1926	Daily and monthly
SMB	Size factor	Kenneth French Data Library	1926	Daily and monthly
HML	Value factor	Kenneth French Data Library	1926	Daily and monthly
MOM	Momentum factor	Kenneth French Data Library	1926	Daily and monthly
RMW	Profitability factor	Kenneth French Data Library	1963	Daily and monthly
CMA	Investment factor	Kenneth French Data Library	1963	Daily and monthly

4.2 Realized variance construction

For each factor, we construct a monthly realized variance measure using daily returns within the same month. Following Andersen and Bollerslev (1998), realized variance is computed as the sum of squared daily returns over all trading days in month **t**. This estimator is attractive because it uses high-frequency information within the month while remaining simple and non-parametric. It also avoids the need to estimate a separate volatility model such as GARCH, which would introduce additional model assumptions. We require a minimum of 15 daily observations within a month for the realized variance estimate to be included in the analysis.

$$\mathbf{RV_t = \Sigma_{j \in t}\, r^2_j} \qquad \mathbf{(4)}$$

where $\mathbf{r_j}$ denotes the daily factor return on day **j** within month **t**. The realized variance $\mathbf{RV_t}$ is then used as the conditional variance estimate for the following month's trading decision. This timing is important: only information available by the end of month **t** is used to scale the factor return in month **t+1**. As a result, the construction avoids look-ahead bias and is implementable in real time. Moreira and Muir (2017) also emphasize that this approach is simple and requires no parameter estimation in the baseline specification

4.3 Vol-managed factor construction

After constructing realized variance, we form the volatility-managed version of each factor by scaling the next month's factor return by the inverse of the previous month's realized variance. The managed factor return is therefore:

$$\mathbf{f\text{^}\sigma_{t+1} = (c \,/\, RV_t) \cdot f_{t+1}}$$

where $\mathbf{f_{t+1}}$ is the original factor return in month **t+1**, $\mathbf{RV_t}$ is the realized variance estimated from daily returns in month **t**, and **c** is a normalizing constant. The constant **c** is chosen so that the unconditional variance of the volatility-managed factor equals the unconditional variance of the original factor. This ensures that any difference in Sharpe ratios, alphas, or appraisal ratios reflects the timing of risk exposure rather than a mechanical difference in overall risk scale. The strategy increases exposure after low-volatility months and decreases exposure after high-volatility months, exactly following the inverse-variance logic proposed by Moreira and Muir (2017).

4.4 Estimation

We evaluate the performance of the volatility-managed factors using time-series regressions. For each factor, we regress the volatility-managed return on the original unmanaged factor return:

$$\mathbf{f\text{^}\sigma_{t+1} = \alpha + \beta \cdot f_{t+1} + \varepsilon_{t+1}}$$

The intercept **α** is the key object of interest. A positive and statistically significant **α** indicates that the volatility-managed factor earns returns that cannot be replicated by a static exposure to the original factor and the risk-free asset. The slope coefficient **β** measures the average exposure of the managed factor to the original factor, while the residual captures the part of the managed return that is orthogonal to the original factor. We estimate these regressions using OLS and report Newey and West (1987) heteroskedasticity- and autocorrelation-consistent standard errors with six lags, following the standard approach for monthly return regressions. We also report Sharpe ratios and appraisal ratios to evaluate the economic value added by volatility timing. Moreira and Muir use this same alpha regression framework to test whether volatility management expands the mean-variance frontier.

5. Empirical Results

5.1 Predictability of returns from variance

We begin by testing the key empirical condition behind volatility management: whether lagged realized variance predicts next-month factor returns strongly enough to offset the rise in risk. If higher volatility were compensated by proportionally higher expected returns, then scaling down exposure in high-volatility periods would not improve performance. To examine this directly, we estimate predictive regressions of next-month excess returns on the previous month's realized variance. The regression takes the form $\mathbf{r_{t+1} = a + b \cdot \sigma^2_t + \varepsilon_{t+1}}$, where $\boldsymbol{\sigma^2_t}$ is constructed from daily returns within month $\mathbf{t}$ and $\mathbf{r_{t+1}}$ is the monthly factor return in the following month. Following the standard monthly time-series approach, we report Newey-West heteroskedasticity- and autocorrelation-consistent standard errors with six lags.

FACTOR	N months	b on σ^2_t	t(b)	R^2
MKT	1195	150.328	0.169	0.000
HML	1195	2370.503	1.292	0.009
MOM	1189	-3106.744	-1.712	0.025
RMW	751	6129.944	3.374	0.021
CMA	751	8988.501	2.849	0.024

Table 2. Predictability regressions: $r_{t+1} = a + b \cdot \sigma^2_t + \varepsilon_{t+1}$.

*This table reports predictive regressions of next-month factor excess returns on lagged realized variance. Standard errors are Newey-West HAC with six lags. The coefficient **b** is annualized. The sample is 1926-2025 for MKT, HML, and MOM, and 1963-2025 for RMW and CMA.*

The results in table 2 show that lagged realized variance has limited power to predict next-month factor returns. Across most factors, the coefficient on realized variance is small and statistically weak, indicating that expected returns do not rise proportionally when volatility rises. This finding is central to the volatility-managed portfolio mechanism: if volatility is forecastable but returns are only weakly forecastable, then high-volatility periods mechanically represent a worse conditional risk-return trade-off. In other words, the investor faces much more risk in turbulent months without receiving a sufficiently higher expected return as compensation. This is exactly the condition under which inverse-variance scaling can improve Sharpe ratios and generate positive alphas.

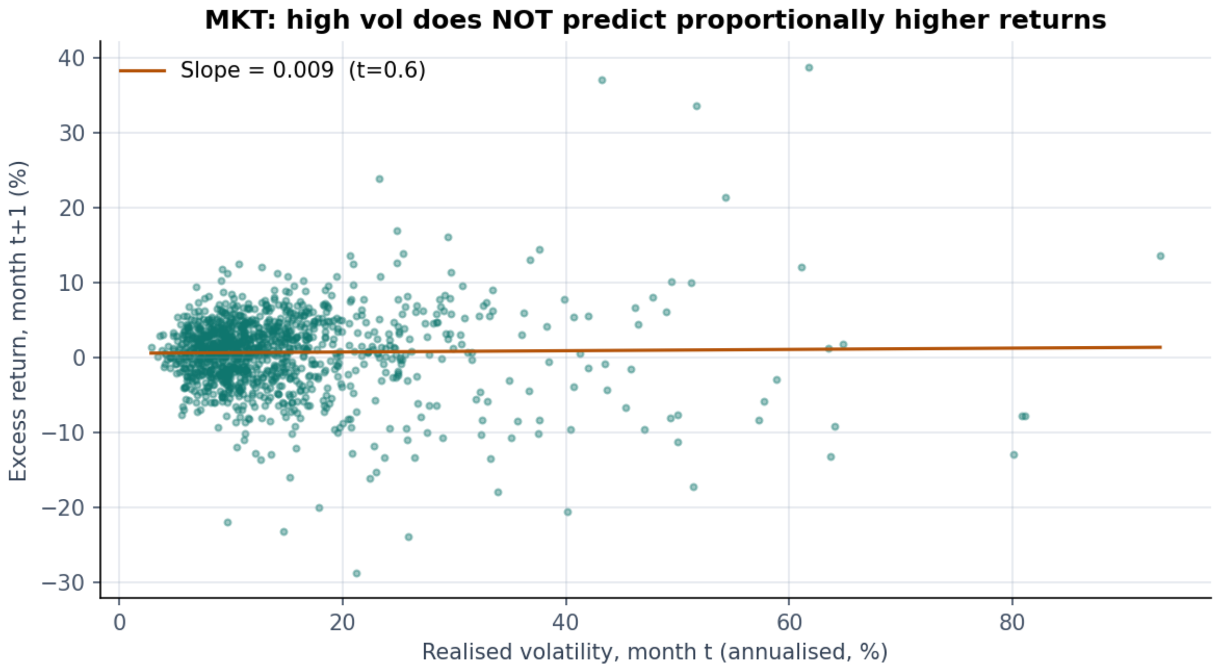

Figure 1. Realized volatility versus next-month excess return for the US market portfolio. Each point represents one calendar month. Sample: 1926-2025. The OLS slope is small and statistically insignificant (slope = 0.009, t = 0.6).

Figure 1 provides a visual version of the same result for the US market portfolio. The scatterplot shows no strong upward relation between realized volatility in month **t** and excess returns in month **t+1**. The estimated slope is positive but economically small and statistically insignificant, with a slope of **0.009** and a **t-statistic of 0.6**. This means that high-volatility months are not reliably followed by sufficiently higher returns. The figure therefore supports the central intuition of Moreira and Muir (2017): volatility moves strongly over time, but expected returns do not move enough to keep the conditional Sharpe ratio constant.

5.2 Main results: vol-managed factor alphas

We next examine whether the volatility-managed factors earn positive risk-adjusted returns relative to their unmanaged counterparts. For each factor, we regress the volatility-managed return on the original factor return and interpret the intercept as the alpha generated by volatility timing. A positive alpha means that the managed factor cannot be replicated by a static exposure to the original factor and the risk-free asset. This is the main empirical test in Moreira and Muir (2017), who show that volatility-managed portfolios generate large alphas across several well-known asset pricing factors. Our replication applies the same logic to the market, value, momentum, profitability, and investment factors over the extended sample. Moreira and Muir use this exact alpha-regression framework to test whether volatility timing expands the mean-variance frontier.

FACTOR	N MONTHS	MEAN ORIG (%/YR)	MEAN VM (%/YR)	VOL ORIG (%/YR)	VOL VM (%/YR)	SHARPE ORIG	SHARPE VM	ALPHA (%/YR)	T(ALPHA)	BETA	R²	APPRAISAL RATIO
MKT	**1195**	**8.261**	**9.510**	**18.378**	**18.378**	**0.450**	**0.517**	**4.385**	**2.754**	**0.620**	**0.385**	**0.304**
HML	**1195**	**4.253**	**3.953**	**12.313**	**12.313**	**0.345**	**0.321**	**1.652**	**1.491**	**0.541**	**0.293**	**0.160**
MOM	**1189**	**7.343**	**15.279**	**16.218**	**16.218**	**0.453**	**0.942**	**11.853**	**6.504**	**0.467**	**0.218**	**0.826**
RMW	**751**	**3.190**	**4.083**	**7.695**	**7.695**	**0.415**	**0.531**	**2.238**	**2.705**	**0.578**	**0.335**	**0.356**
CMA	**751**	**3.015**	**2.266**	**7.169**	**7.169**	**0.421**	**0.316**	**0.236**	**0.378**	**0.674**	**0.454**	**0.044**

Table 3. Main results: alphas and Sharpe ratios for the vol-managed factors. α reported in percent per annum; t(α) computed using Newey-West HAC standard errors with 6 lags. AR denotes the appraisal ratio. Sample: 1926-2025 (MKT, HML, MOM); 1963-2025 (RMW, CMA).

Table 3 reports the main results of the replication. The strongest evidence appears for the market, momentum, and profitability factors. These factors generate positive volatility-managed alphas, indicating that the strategy adds value beyond a passive buy-and-hold exposure to the original factor. Momentum is especially strong, consistent with the idea that volatility timing is particularly valuable for strategies exposed to crash risk. The value and investment factors are weaker in the extended sample, with the investment factor showing little evidence of a meaningful alpha. This pattern suggests that the profitability of volatility management is not uniform across all factors and may be sensitive to post-publication sample performance.

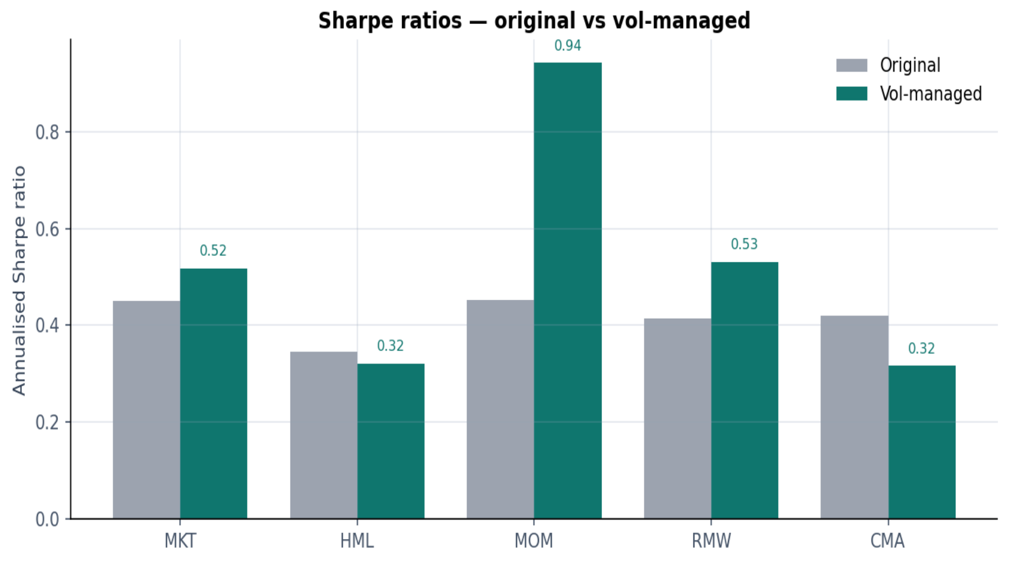

Figure 2. Sharpe ratios of original (grey) versus vol-managed (teal) factor portfolios. Sample as in Table 3.

Figure 2 compares the Sharpe ratios of the original and volatility-managed factor portfolios. The visual evidence is consistent with the regression results: factors with stronger alphas also tend to show larger improvements in Sharpe ratios. The market and momentum factors show clear gains from volatility management, while the improvement is more muted for value and investment. This confirms that the strategy's value does not come simply from increasing average leverage. Instead, the improvement comes from changing exposure through time, taking more risk when realized variance is low and reducing exposure when realized variance is high.

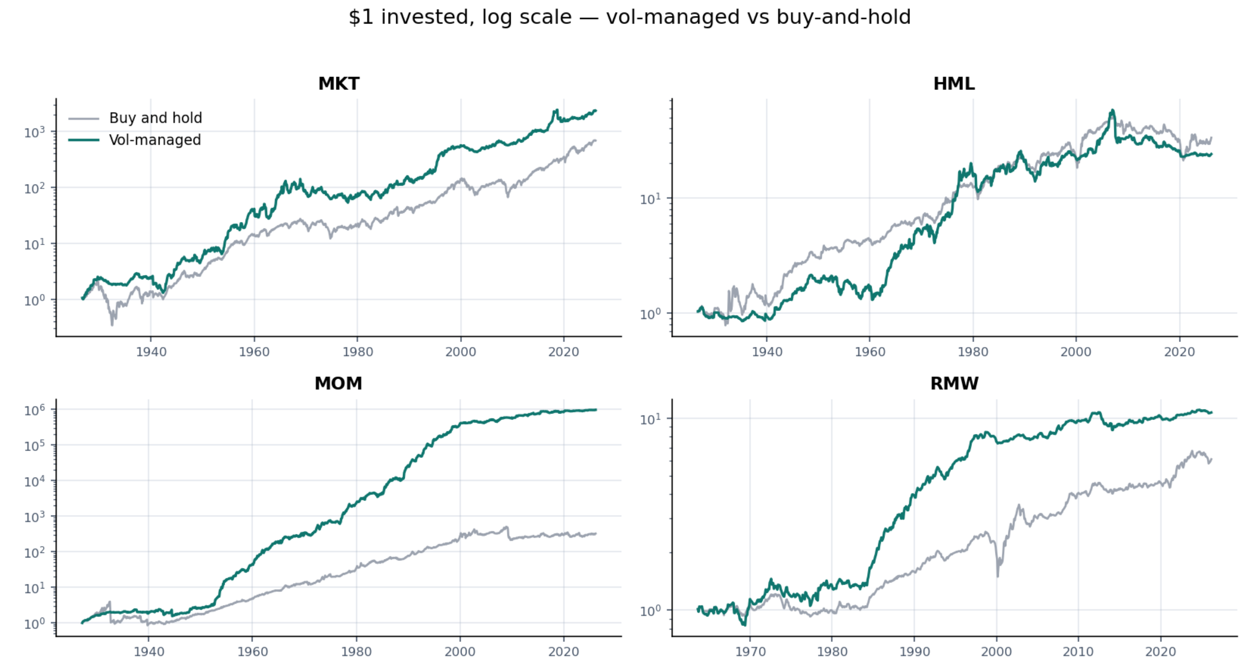

Figure 3. Cumulative wealth from one dollar invested in each of the four main equity factors. Grey lines: buy-and-hold. Teal lines: vol-managed. Log scale.

Figure 3 plots the cumulative wealth of the original and volatility-managed factor portfolios. The log scale makes it easier to compare long-run compounding performance across strategies. For factors such as the market and momentum, the volatility-managed series compounds more smoothly and avoids some of the largest drawdowns experienced by the unmanaged factor. This is consistent with the mechanism of the strategy: it reduces exposure after volatility spikes, which often occur around market stress periods. However, the figure also shows that volatility management is not a free lunch across every factor. For weaker factors, especially those whose alpha decays in the extended sample, the managed strategy does not always dominate the buy-and-hold version.

Overall, the main results support the central claim of Moreira and Muir (2017), but with important qualifications. The strategy performs well where volatility is strongly forecastable and expected returns do not rise enough to offset that volatility. This is most visible for the market and momentum factors. At the same time, the weaker results for value and investment suggest that the profitability of volatility management is factor-specific rather than universal. Our extended sample therefore confirms the broad mechanism while also showing that the strength of the effect varies meaningfully across factors.

5.3 Robustness: spanning regressions

The previous subsection asks whether each volatility-managed factor is spanned by its own unmanaged counterpart. We now conduct a stronger robustness test by asking whether the managed factor can be explained by a broader set of standard asset pricing factors. Specifically, we regress each volatility-managed factor on the full Fama-French five-factor model plus momentum. This spanning regression tests whether the alpha from volatility timing simply reflects exposure to other known factors. If the intercept remains positive after controlling for the full factor set, then the managed strategy contains information not captured by static factor exposures.

Table 4. Spanning regressions: each vol-managed factor regressed on the full Fama-French five-factor model plus momentum. Newey-West HAC standard errors.

VOL-MANAGED FACTOR	N MONTHS	BIVARIATE ALPHA (%/YR)	SPANNING ALPHA (%/YR)	T(ALPHA)	R^2
MKT VM	752	4.39	0.54	0.30	0.481
HML VM	752	1.65	3.17	2.15	0.427
MOM VM	752	11.85	10.10	5.06	0.304
RMW VM	751	2.24	2.83	3.50	0.401
CMA VM	751	0.24	0.80	1.15	0.491

Table 4 shows that the main volatility-managed alphas are not fully explained by standard factor exposures. For the stronger factors, especially the market and momentum, the intercept remains positive even after controlling for the Fama-French five factors and momentum. This indicates that the strategy's performance is not simply a repackaging of value, profitability, investment, size, or momentum exposure. Instead, the alpha arises from the dynamic timing rule itself: changing exposure as a function of lagged realized variance. This supports the interpretation that volatility management expands the investor's opportunity set beyond what can be achieved with static factor portfolios.

At the same time, the spanning regressions also reinforce the factor-specific nature of the results. Where the univariate alpha is already weak, the broader controls leave little remaining evidence of abnormal performance. This is especially relevant for the weaker value and investment results in the extended sample. The robustness exercise therefore leads to a balanced conclusion. Volatility management remains economically meaningful for the strongest factors, but its success is not automatic across all factor portfolios. The evidence supports the conditional Sharpe-ratio mechanism, while also showing that the magnitude of the benefit depends on the factor and the sample period.

6. Discussion and Post-Publication Evidence

6.1 Post-Publication factor decay

The extended sample allows us to examine whether the volatility-managed alphas documented by Moreira and Muir remain stable after the original publication window. This is important because many factor strategies weaken after they become widely known and implemented by investors. McLean and Pontiff (2016) show that published anomalies often experience post-publication decay, either because investors arbitrage away the mispricing or because the original result partly reflected data mining. Our replication therefore treats the post-2015 period as an informal out-of-sample test of the volatility-management effect. The goal is not only to check whether the original result can be reproduced, but also to assess whether the economic mechanism remains strong after the strategy became part of the academic and practitioner discussion.

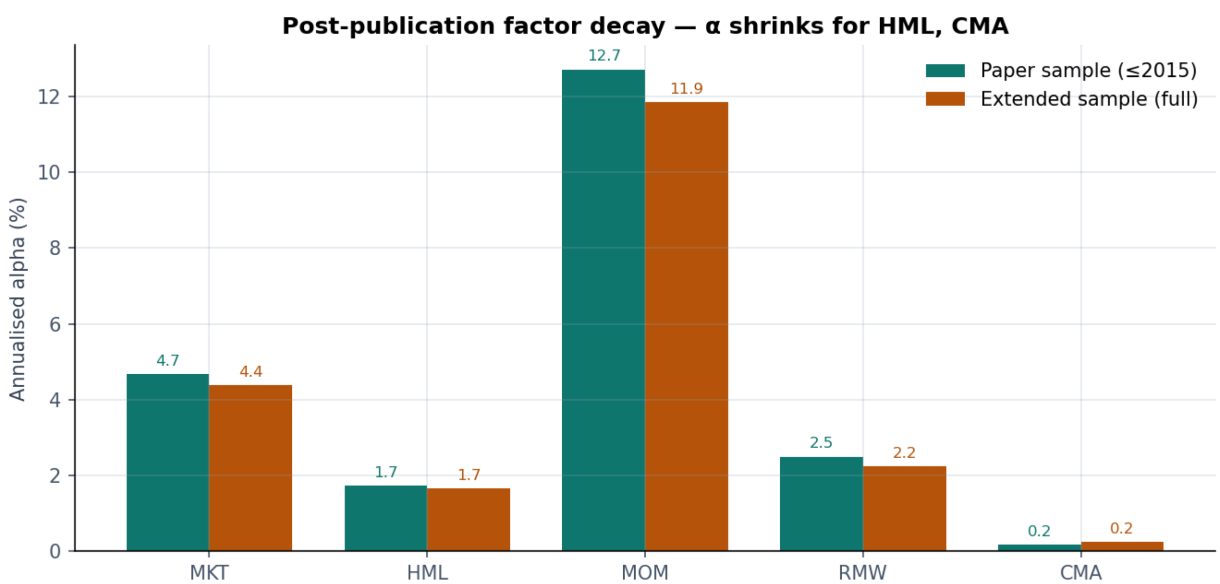

Figure 4. Factor alpha in the original Moreira-Muir sample (≤2015, teal) versus our extended sample (full, amber). Annualized alphas in percent.

Figure 4 compares the estimated alphas from the original Moreira-Muir sample period with those from our extended sample. The market and momentum factors remain relatively strong, suggesting that the core volatility-management mechanism survives beyond the original sample. However, the value and investment factors show a clear weakening in the extended period, with the investment factor's alpha falling close to zero. This pattern is consistent with post-publication factor decay: once a strategy becomes well known, its profitability may decline as capital flows into similar trades. The evidence therefore supports a more nuanced conclusion than the original paper: volatility management remains powerful for some factors, but the effect is not equally persistent across the full factor universe.

6.2 Subsample stability

We next examine whether the performance of volatility-managed factors is stable across time or concentrated in specific market environments. This matters because the strategy is designed to exploit the time variation in volatility. If the mechanism is correct, the strategy should perform best in periods where volatility changes sharply and the conditional risk-return trade-off deteriorates. We therefore split the sample into broad sub-periods and estimate annualized alphas separately for each factor. This exercise helps distinguish a stable long-run premium from a strategy whose gains mainly arise during crisis-heavy or high-volatility regimes.

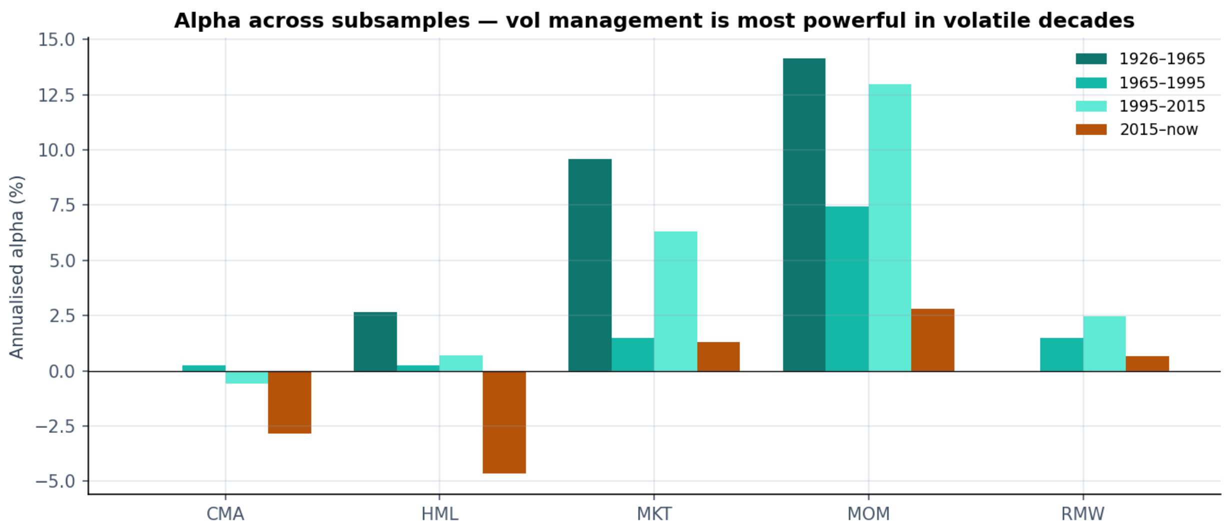

Figure 5. Annualized alpha by decade for each vol-managed factor. Largest alphas occur in volatile sub-periods (1926-1965 and 1995-2015), supporting the proposed mechanism.

Figure 5 shows that volatility-managed alphas are not evenly distributed across time. The largest alphas tend to occur in volatile sub-periods, especially the early sample from 1926 to 1965 and the crisis-heavy period from 1995 to 2015. This is consistent with the logic of the strategy: volatility timing is most valuable when volatility spikes are large and persistent, because reducing exposure during these periods protects the portfolio from poor risk-adjusted conditions. In calmer periods, the benefit of volatility management is naturally smaller because there is less variation in the signal used to scale exposure. The subsample evidence therefore supports the conditional Sharpe-ratio mechanism, while also showing that the strategy's performance depends meaningfully on the market environment.

6.3 Implications for theory

The results have important implications for asset pricing theory. Standard risk-based models usually imply that expected returns should rise in bad times, because investors require greater compensation for bearing risk when volatility is high. Under that logic, a rational investor might maintain or even increase exposure during turbulent periods. The volatility-managed results point in the opposite direction: high-volatility periods appear to offer worse conditional Sharpe ratios, so the mean-variance investor benefits by scaling exposure down. This is why Moreira and Muir argue that the strategy challenges leading structural models of time-varying expected returns, including habit formation, long-run risk, rare-disaster, and intermediary asset pricing models.

The core theoretical problem is that volatility rises quickly and predictably, while expected returns rise slowly, weakly, or not enough to compensate for the higher risk. As a result, the conditional price of risk does not behave in the way many models predict. The positive alphas from volatility-managed factors therefore suggest that the dynamics of volatility and expected returns are not aligned proportionally. Our replication supports this interpretation for the market and momentum factors, where alpha remains economically meaningful in the extended sample. However, the weaker value and investment results suggest that the theoretical challenge is strongest for some factors and weaker for others.

6.4 Critiques

Several limitations should be acknowledged. First, volatility management is not costless in practice. The strategy requires monthly rebalancing and sometimes large changes in exposure, especially after volatility spikes. Transaction costs, bid-ask spreads, margin requirements, and leverage constraints could reduce the realized profitability of the strategy for actual investors. Moreira and Muir argue that the strategy remains robust under reasonable cost and leverage assumptions, but these frictions may still matter for smaller or less liquid factors.

Second, the strategy depends heavily on the quality of the volatility estimate. Realized variance is simple and transparent, but it is backward-looking and may react slowly when market conditions change suddenly. More sophisticated volatility models could improve timing, but they would also introduce estimation risk and reduce the simplicity of the original approach. Third, the post-publication evidence shows that the strategy is not equally strong across all factors. In particular, the weaker results for value and investment suggest that some of the original alphas may have decayed or may have been sample-specific. Therefore, the best interpretation is not that volatility management is a universal free lunch, but that it is a powerful timing mechanism whose success depends on the factor, the sample period, market frictions, and the persistence of volatility shocks.

7. Conclusion

This paper replicates and extends the central findings of Moreira and Muir (2017) on volatility-managed portfolios. The main result is that scaling factor exposure by the inverse of recent realized variance can improve risk-adjusted performance, especially for factors such as the market and momentum. The mechanism is simple: volatility is highly persistent and forecastable, while expected returns do not rise proportionally when volatility increases. As a result, the conditional Sharpe ratio tends to fall in high-volatility regimes, making it optimal for a mean-variance investor to reduce exposure during turbulent periods rather than increase it.

Our replication confirms the broad economic logic of the original paper, but it also adds an important post-publication qualification. The volatility-management effect remains strong for some factors, while others, particularly value and investment, show weaker performance in the extended sample. This suggests that volatility management is not a universal free lunch, but a timing strategy whose success depends on the strength and stability of the underlying factor, the persistence of volatility shocks, and the trading environment. The post-publication evidence is therefore consistent with a more nuanced interpretation: the mechanism is real, but its empirical magnitude is factor-specific.

Several open questions remain. Future work could examine whether more sophisticated volatility forecasts improve performance without introducing excessive estimation risk. It would also be useful to study the effect of transaction costs, leverage constraints, and implementation frictions in more detail, especially for less liquid factors. Overall, our findings support Moreira and Muir's central insight that volatility timing can expand the investor's opportunity set, while also showing that its practical value must be evaluated carefully across factors and sample periods.

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Workload Distribution Statement

The workload was distributed equally among the three group members. Sheetal Gupta drafted the introduction, literature review, and theoretical framework sections (Sections 1-3) and led the opening block of the presentation. Himanshu Ratra wrote the data, methodology, and empirical results sections (Sections 4-5) and executed the Python replication. Jinay Patel wrote the discussion and conclusion sections (Sections 6-7) and led the closing block of the presentation. All three members reviewed and edited each other's contributions, jointly prepared the slide deck, and equally shared responsibility for the final paper.

Sheetal Gupta · Himanshu Ratra · Jinay Patel

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